Assignment: Ecommerce platform

Business Case

An e-commerce marketplace connecting sellers to customers.

Management has the following strategic question:

“What drives revenue and customer lifetime value, and how are delivery performance and customer satisfaction impacting long-term business performance across products and regions?”

Technical Requirements

You must implement a medallion architecture:

/Raw
/Bronze
/Silver
/Gold

Technology

- Google Colab
- PySpark
- Spark SQL
- Plots via Spark / Pandas / Matplotlib / Seaborn

Source Files

Use at minimum:

- orders
- order_items
- order_payments
- order_reviews
- customers
- products
- sellers
- geolocation
- product_category_translation

You may justify excluding files if properly explained.

Bronze Layer (Raw Ingestion)

Requirements:

- Load all CSVs as-is
- No schema modification
- Store in Bronze as parquet
- Add ingestion timestamp
- No transformations allowed

Show:

- Schema
- Row counts
- Example records

Silver Layer (Cleansed & Standardized)

✓ Data Cleaning

- Handle nulls
- Standardize column naming (snake_case)
- Convert timestamps to proper types
- Remove duplicates

✓ Data Standardization

- Fix inconsistent state codes
- Normalize text casing
- Convert monetary columns to numeric

✓ Implement SCD Type 2

You must implement SCD Type 2 for:

- Customers (location changes)
- Sellers (if applicable)

Include:

- surrogate_key
- valid_from
- valid_to
- is_current flag

✓ Keys

- Generate surrogate keys for all dimensions
- Keep business keys

Gold Layer (Business Model – Star Schema)

You must model a dimensional star schema.
Required Dimensions

- dim_customer (SCD2)
- dim_product
- dim_seller
- dim_date
- dim_geography

Required Fact Table

fact_sales

Must include:

- order_id
- product_key
- customer_key
- seller_key
- date_key
- revenue
- freight_value
- review_score
- delivery_delay_days
- order_status

Upsert Requirement

Gold tables must be written using MERGE (Upsert) logic.

No full overwrite allowed.

Required Business Insights (7 Mandatory Visualizations)

1️⃣ Revenue by Product Category

(All categories)

2️⃣ Top 10 Customers by Lifetime Revenue

(Use window functions)

3️⃣ Revenue vs Average Review Score

(Correlation analysis)

4️⃣ On-Time vs Late Deliveries (%) per State

5️⃣ Top 10 Cities by Revenue

6️⃣ Monthly Revenue Trend (Time Series)

7️⃣ Customer Lifetime Value Distribution

(Segmentation: Low / Medium / High using CASE)

Advanced Analytical Requirements

✓ Create Customer Segmentation

Using:

- Total revenue
- Number of orders
- Average review score

Segment into:

- VIP
- Loyal
- At Risk
- One-time buyers

Justify segmentation logic.

✓ Compute Delivery Performance KPI

delivery_delay_days = actual_delivery_date - estimated_delivery_date

Classify:

- Early
- On-time
- Late

✓ Use at least 2 Window Functions

Examples:

- RANK()
- DENSE_RANK()
- SUM() OVER()
- ROW_NUMBER()

Deliverables

One Colab Notebook (.ipynb)

Must contain:

- Clear comments per layer
- Explanation of modelling decisions
- Screenshots of tables per layer
- SCD logic explanation

Seven (7) Visualizations

Clear, readable, business-focused.

PowerPoint Presentation

Must include:

1. Business understanding
2. Architecture design (Bronze/Silver/Gold)
3. Star schema diagram
4. SCD explanation
5. 7 visualizations
6. Final management insights